

Syllabus : Determinants, Area Under Curve

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- If $f(x)$, $g(x)$ and $h(x)$ are three polynomials of degree 2 and $\Delta(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}$ then $\Delta'(x)$ is a polynomial of degree (whenever defined)

(1) 2 (2) 3 (3) 0 (4) 1
- The equations $kx + y + z = k - 1$, $x + ky + z = k + 1$, $x + y + kz = k - 1$ has no solution for

(1) $k = 1$ only (2) $k \neq -2$ (3) $k = -2$ or 1 (4) $k = -2$ only
- If $\Delta = \begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$ then

(1) $\Delta_{\max} = 10$ (2) $\Delta_{\max} = \sqrt{10}$ (3) $\Delta_{\min} = 10$ (4) $\Delta_{\min} = -10$
- The system of equations in x, y, z (all real) given as $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$, $\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ and $\frac{-x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ has ($a, b, c \in \mathbb{R}$)

(1) no solution (2) unique solution
 (3) infinitely many solutions (4) finitely many solutions
- Let p, q, r be real numbers such that $p + q + r \neq 0$. The system of linear equations

$$\begin{aligned} x + 2y - 3z &= p \\ 2x + 6y - 11z &= q \\ x - 2y + 7z &= r \end{aligned}$$

has at least one solution if :

(1) $5p + 2q - r = 0$ (2) $5p - 2q - r = 0$ (3) $5p + 2q + r = 0$ (4) $5p - 2q + r = 0$
- The system of linear equations

$$\begin{aligned} x - y + z &= 1 \\ x + y - z &= 3 \\ x - 4y + 4z &= \alpha \end{aligned}$$

has :

(1) a unique solution when $\alpha = 2$ (2) a unique solution when $\alpha \neq 2$
 (3) an infinite number of solutions, when $\alpha = 2$ (4) an infinite number of solutions, when $\alpha = -2$

7. If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 23 & 17 & 13 \\ 1 & 1 & 1 \end{vmatrix}$, $x \in \mathbb{R}$, then $\frac{d^2 y}{dx^2} + y$ is equal to :
- (1) 6 (2) 4 (3) -10 (4) 0
8. Let $\{\Delta_1, \Delta_2, \dots, \Delta_n\}$ be the set of all determinants of order 3 that can be made with the distinct real numbers from the set $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Then which of the following is correct?
- (1) $\sum_{i=1}^n \Delta_i = 0$ (2) $\sum_{i=1}^n \Delta_i = 9$ (3) $\sum_{i=1}^n \Delta_i = 9!$ (4) $\sum_{i=1}^n \Delta_i = 36!$
9. Let $A = \begin{bmatrix} \beta & -1 \\ 1 & 2\beta \end{bmatrix}$ and $\det(A^4) = 16$ then the product of all possible real values of β is
- (1) $\frac{1}{2}$ (2) $-\frac{1}{2}$ (3) 0 (4) 2
10. If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)(x-1) \end{vmatrix}$ then $f(100) =$
- (1) 0 (2) 100 (3) 1 (4) None of these
11. If the system of equations $x + ay = 0$, $az + y = 0$, $ax + z = 0$ has infinite number of solutions, then $a =$
- (1) 0 (2) 1 (3) -1 (4) -2
12. If $S = \left\{ x \in [0, 2\pi] : \begin{vmatrix} 0 & \cos x & -\sin x \\ \sin x & 0 & \cos x \\ \cos x & \sin x & 0 \end{vmatrix} = 0 \right\}$ then $\sum_{x \in S} \tan\left(\frac{\pi}{3} + x\right)$ is equal to
- (1) $-2 + \sqrt{3}$ (2) $-2 - \sqrt{3}$ (3) $-4 - 2\sqrt{3}$ (4) $4 + 2\sqrt{3}$
13. If the system of linear equations $x + 2ay + az = 0$, $x + 3by + bz = 0$ and $x + 4cy + cz = 0$ has a non-zero solution, then a, b, c
- (1) are in AP (2) are in GP (3) are in HP (4) satisfy $a + 2b + 3c = 0$
14. Let $N = \begin{vmatrix} 28 & 25 & 38 \\ 42 & 38 & 65 \\ 56 & 47 & 83 \end{vmatrix}$ then the number of ways in which N can be resolved as a product of two divisors which are relatively prime is
- (1) 4 (2) 8 (3) 9 (4) 16
15. If $x = a$, $y = b$, $z = c$ is a solution of the system of linear equations $x + 8y + 7z = 0$, $9x + 2y + 3z = 0$, $x + y + z = 0$ such that the point (a, b, c) lies on the plane $x + 2y + z = 6$, then $2a + b + c$ equals
- (1) 2 (2) 1 (3) 0 (4) -1
16. Let area enclosed by curve $y = 1 - x^2$ and $y \geq a$ where $0 \leq a < 1$ is $A(a)$. If $A(0) = kA(1/2)$ then
- (1) $k = 0$ (2) $0 < k < 1$ (3) $1 < k < 2$ (4) $2 < k < 3$
17. The area bounded by curves $y = \ln x$ and $y = (\ln x)^2$ is
- (1) $e - 1$ (2) $e - 2$ (3) $3 - e$ (4) e
18. Area bounded by curves $1 - x = 5y^2$ and $-4y^2 = x$ is
- (1) 1 (2) 2 (3) $2/3$ (4) $4/3$
19. The area of the region bounded by the curves $y = 1 - x^2$, $x + y + 1 = 0$ and $x - y - 1 = 0$ is -
- (1) 3 (2) $10/3$ (3) $7/3$ (4) $8/3$
20. If the line $y = mx$, bisects the area of the region $\{(x, y) : 0 \leq x \leq \frac{3}{2}, 0 \leq y \leq 1 + 4x - x^2\}$, then m equals
- (1) $\frac{39}{16}$ (2) $\frac{9}{8}$ (3) $\frac{13}{3}$ (4) $\frac{13}{6}$

SECTION-II (NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. Find the number of the values of k for which the system of equations $x + y + z = 1$, $x + 2y + 4z = k$, $x + 4y + 10z = k^2$ has a solution.
22. Consider a determinant $D = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$ where $a, b, c, d \in \{0, 1\}$. If n denotes the total number of determinants whose entries are 0 or 1 only and m denotes the number of determinants with non-zero value, then the ratio $\frac{m}{n}$ is equal to k then find value of $[100k]$, $[.]$ is greatest integer function
23. Find the number of solutions of equation $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 + \sin \theta & 1 \\ 1 & 1 & 1 + \cot \theta \end{vmatrix} = 0$ in $\theta \in [0, 2\pi]$
24. What is the area enclosed in square units by curve $y^2 = -x|x|$ and $x = -1$?
25. The area bounded by $y = x e^{|x|}$ and the lines $|x| = 1$, $y = 0$ is -

ANSWER KEY_DPP-17

1.	(3)	2.	(3)	3.	(1)	4.	(4)	5.	(2)	6.	(4)	7.	(1)
8.	(1)	9.	(2)	10.	(4)	11.	(3)	12.	(3)	13.	(3)	14.	(2)
15.	(2)	16.	(4)	17.	(3)	18.	(4)	19.	(3)	20.	(4)	21.	(2)
22.	(37)	23.	(2)	24.	(1)	25.	(2)						